Chebyshev Quadrature Versus Trapezoid Integration

Scalar, random-polynomial, and EuRoC aggressive-snippet evidence

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1 Executive Summary

The evidence supports a conditional, not universal, case for Chebyshev2 quadrature. The corrected Chebyshev2 integration path now evaluates the antiderivative in an `N+1` representation, and the regularized runs use the derivative-matrix Tikhonov term with `lambda1 = 0.005`. Those fixes make the comparison fairer, but they do not make Chebyshev2 dominate trapezoid integration in every regime.

The consistent pattern is:

- Chebyshev2 is strongest when the signal is smooth and the chosen `N` is modest, usually near or below `sqrt(m)`.
- Trapezoid remains a strong baseline because the samples are uniformly spaced and the data are noisy. It is local and less exposed to global fit artifacts.
- Tikhonov regularization helps most when the Chebyshev2 fit would otherwise use too much high-order freedom. It can improve robustness, but it is still a modeling choice rather than a guaranteed win.
- The EuRoC snippets are the most relevant evidence. They show that Chebyshev2 can win on aggressive 200 ms windows, but the advantage depends on `N`, noise level, and dataset.

Summary over the decision-view comparisons:

Family	lambda1	Overall RMSE win rate	Median RMSE advantage	Mean robust-N win rate	Robust rows with positive advantage
analytic scalar functions	0	63%	0.0001375	85%	100%
analytic scalar functions	0.005	65%	0.0001984	88%	100%
random degree-4 polynomials	0	58%	5.902e-05	100%	100%
random degree-4 polynomials	0.005	61%	7.695e-05	96%	100%
EuRoC aggressive 200 ms snippets	0	38%	-2.246e-05	68%	87%
EuRoC aggressive 200 ms snippets	0.005	46%	-9.268e-06	71%	95%

How to read the figures: positive advantage means `trapezoid error - Chebyshev2 error > 0`, so Chebyshev2 has lower error. The black diamond marks the best median RMSE `N`, and the dashed vertical line marks `sqrt(m)`.

2 Reproducing the May 24 Scalar Comparison

The May 24 scalar notebook can be reproduced with the current corrected integration code. The two figures below are deliberately representative old-style image plots: they establish continuity with Benjamin's original scalar comparison, but they are not the best way to choose `N` because the sign, noise dependence, and fixed-`N` slices are hard to compare directly.

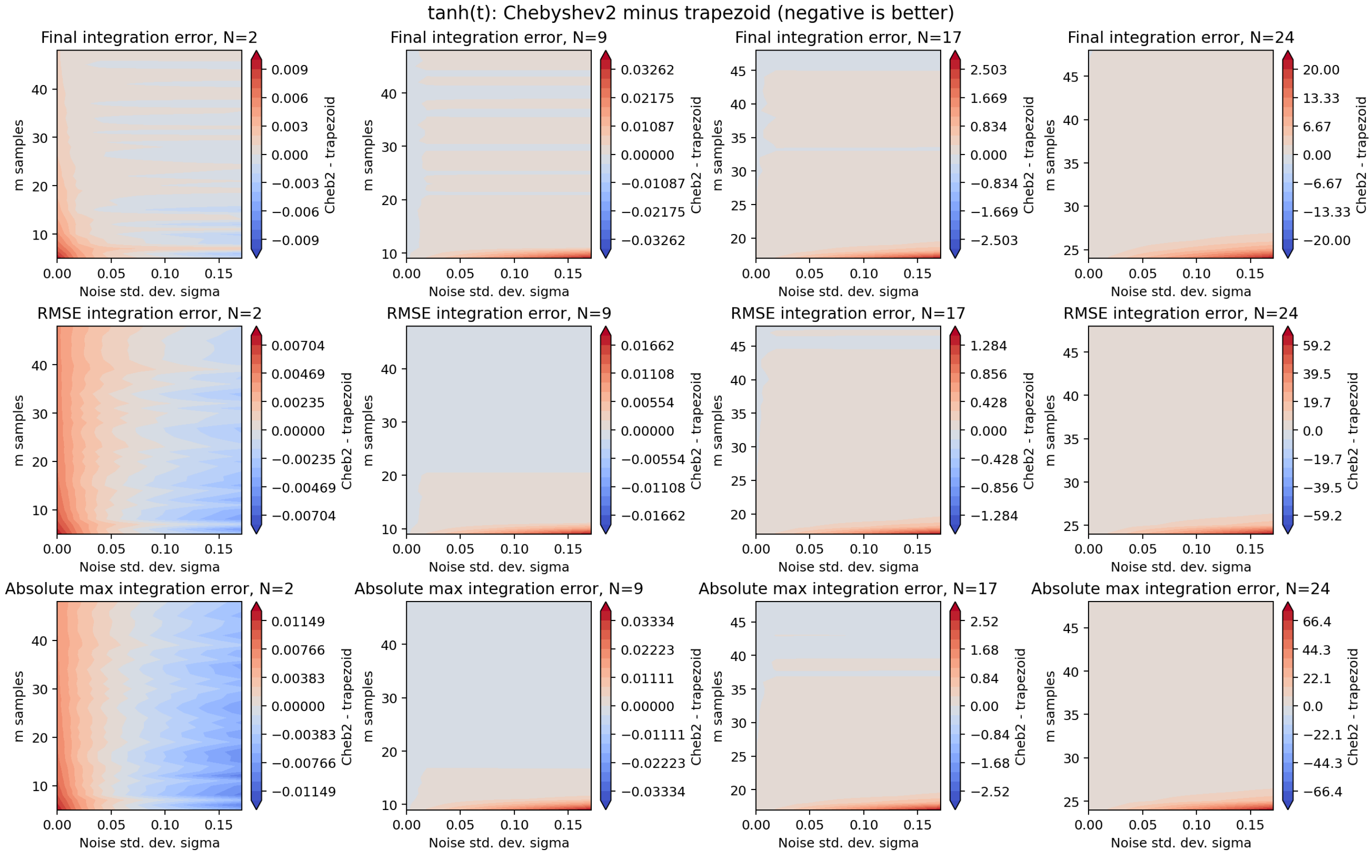

Figure 1: Representative old fixed-N scalar heatmaps for `tanh(t)`.

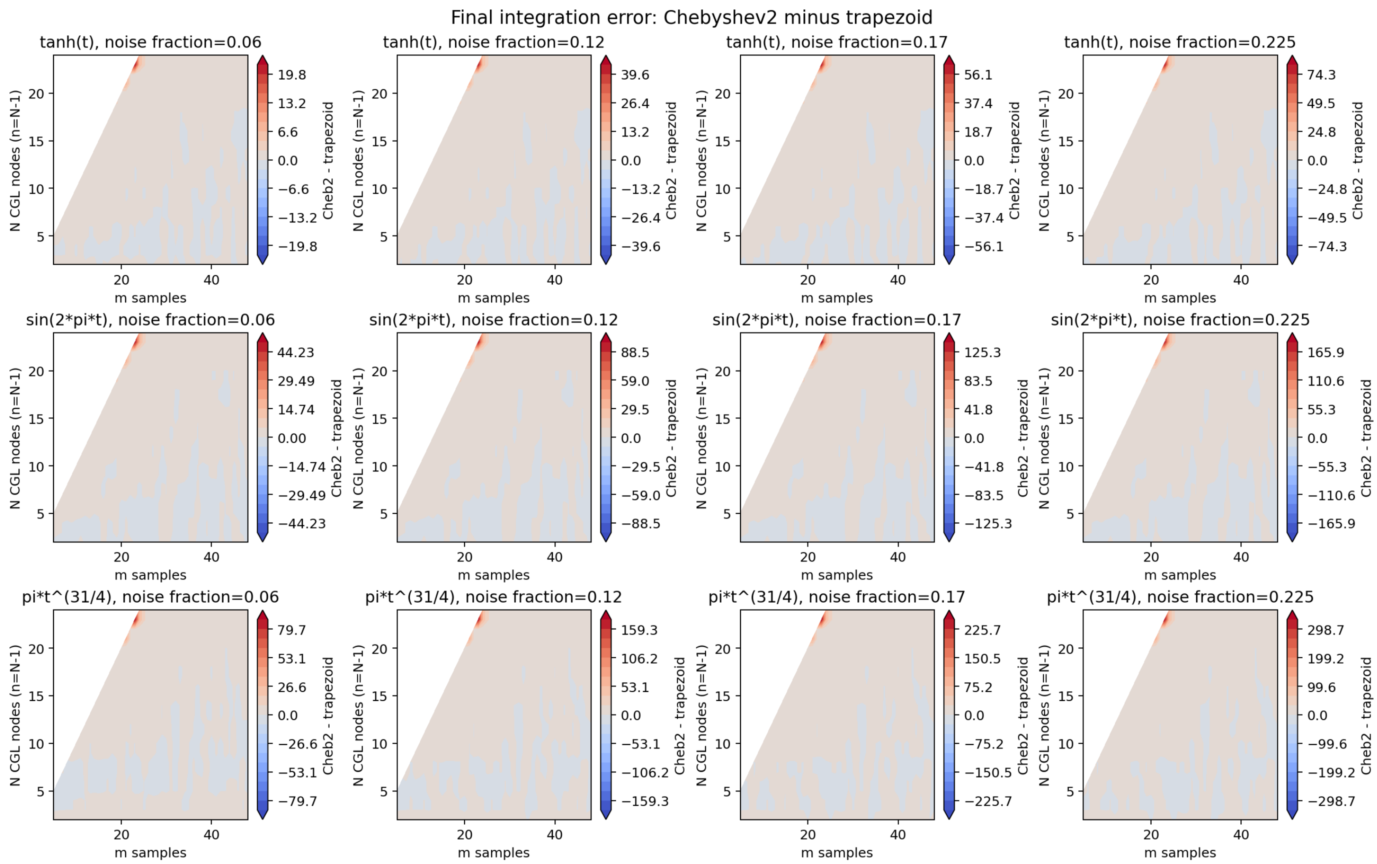

Figure 2: Representative old scalar heatmaps across all three functions at selected noise levels.

3 Decision-oriented analytic scalar views

The same three analytic functions are now shown through the decision-oriented Plotly exports. The unregularized block is the baseline; the regularized block uses lambda1 = 0.005. Positive RMSE advantage means trapezoid error minus Chebyshev2 error is positive, so Chebyshev2 is better.

3.1 No Tikhonov (`lambda1 = 0`)

3.1.1 `tanh(t)`

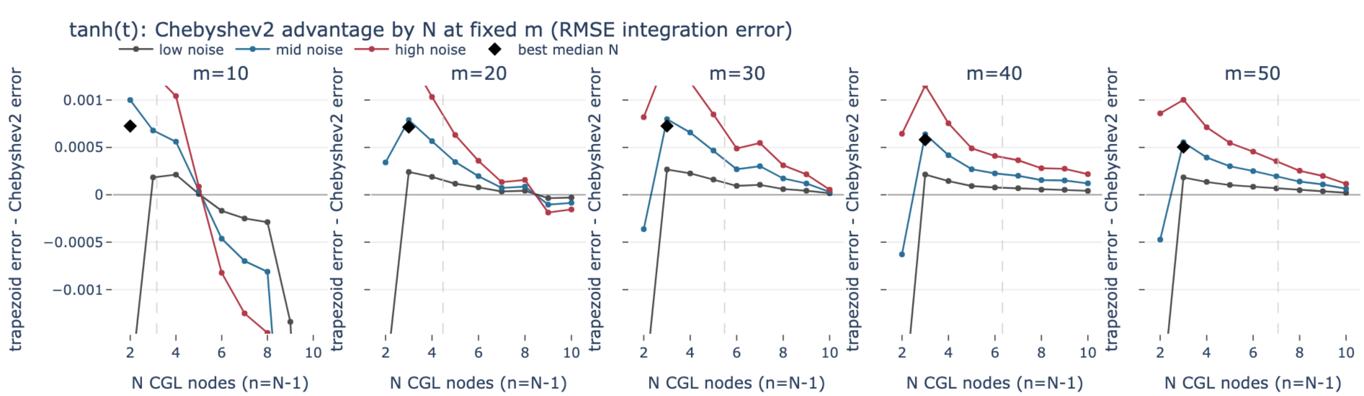

Figure 3: RMSE advantage curves for `tanh(t)` with No Tikhonov (`lambda1 = 0`).

tanh(t): ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	2	2	3	2	0%	75%	100%	0.0007259	@@@@%%%#.
20	4.47	6	3	3	3	75%	100%	100%	0.0007155	:@#+-::..
30	5.48	10	3	3	3	75%	100%	100%	0.0007262	.@%#**++=
40	6.32	9	3	3	3	75%	100%	100%	0.0005816	.@%##****
50	7.07	7	3	3	3	75%	100%	100%	0.000504	.@%##****

Figure 4: Robust `N` table for `tanh(t)` with No Tikhonov (`lambda1 = 0`).

3.1.2 `sin(2*pi*t)`

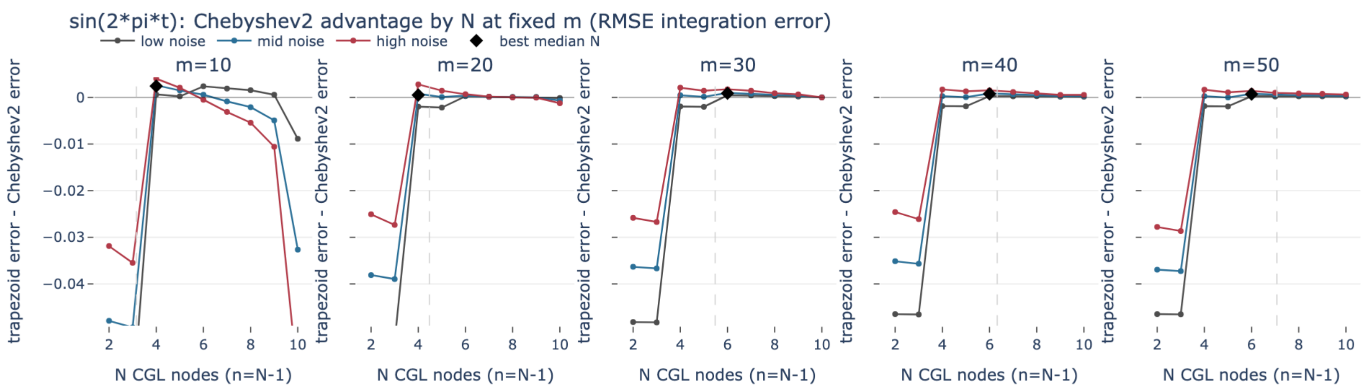

Figure 5: RMSE advantage curves for `sin(2*pi*t)` with No Tikhonov (`lambda1 = 0`).

sin(2*pi*t): ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	2	4	4	4	75%	100%	100%	0.002417	..@@@%%%=
20	4.47	3	4	6	4	0%	75%	100%	0.0005105	..@@@@@@@
30	5.48	10	6	6	6	100%	100%	100%	0.0008826	..@@@@@@@
40	6.32	2	6	6	6	100%	100%	100%	0.000757	..@@@@@@@
50	7.07	2	6	6	6	75%	100%	100%	0.0006953	..@@@@@@@

Figure 6: Robust `N` table for `sin(2*pi*t)` with No Tikhonov (`lambda1 = 0`).

3.1.3 `pi*t^(31/4)`

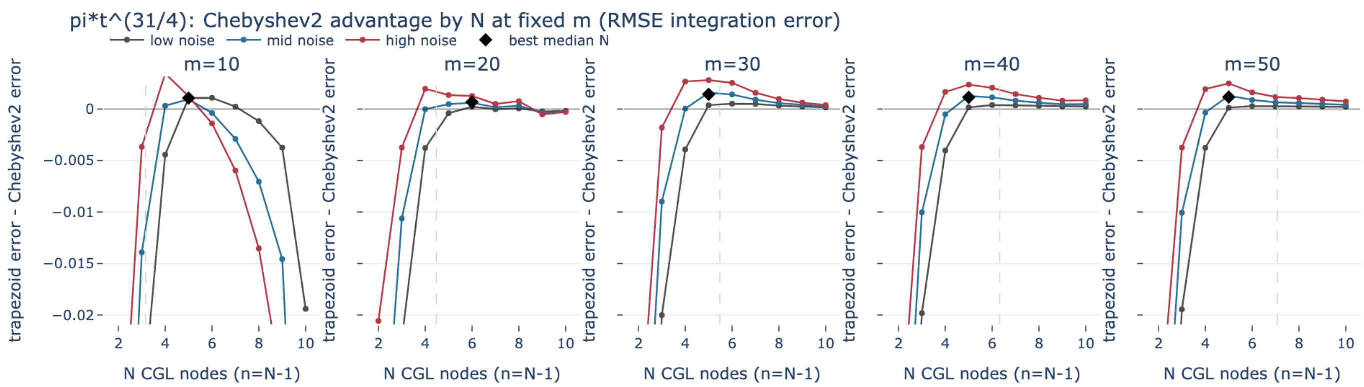

Figure 7: RMSE advantage curves for `pi*t^(31/4)` with No Tikhonov (`lambda1 = 0`).

pi*t^(31/4): ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	5	5	5	5	100%	100%	100%	0.001068	.#@@@@%#_
20	4.47	4	6	5	5	0%	100%	100%	0.0003733	.#@@@@@@@
30	5.48	5	5	5	5	75%	100%	100%	0.001437	.#@@@@@@@
40	6.32	6	5	6	6	75%	100%	100%	0.001035	.#@@@@@@@
50	7.07	9	5	5	5	50%	100%	100%	0.001168	.#@@@@@@@

Figure 8: Robust `N` table for `pi*t^(31/4)` with No Tikhonov (`lambda1 = 0`).

3.2 Tikhonov (`lambda1 = 0.005`)

3.2.1 `tanh(t)`

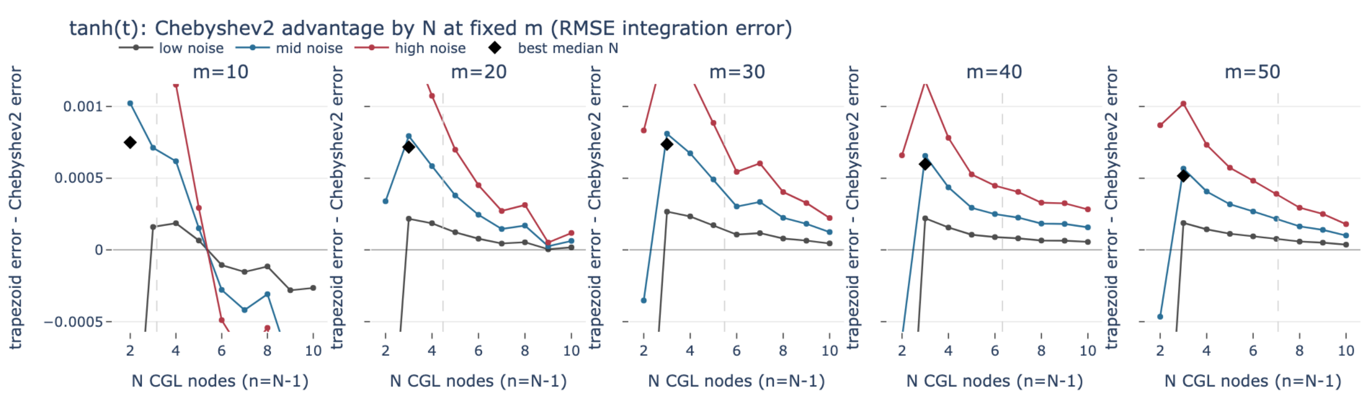

Figure 9: RMSE advantage curves for `tanh(t)` with Tikhonov (`lambda1 = 0.005`).

tanh(t): ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	2	2	3	2	0%	75%	100%	0.0007497	@%%+-:-..
20	4.47	6	3	3	5	75%	100%	100%	0.0003448	_@#=-_:..
30	5.48	9	3	3	3	75%	100%	100%	0.0007368	.@%#**+++
40	6.32	9	3	3	3	75%	100%	100%	0.000598	.@%##****
50	7.07	7	3	3	3	75%	100%	100%	0.0005158	.@%###***

Figure 10: Robust `N` table for `tanh(t)` with Tikhonov (`lambda1 = 0.005`).

3.2.2 `sin(2*pi*t)`

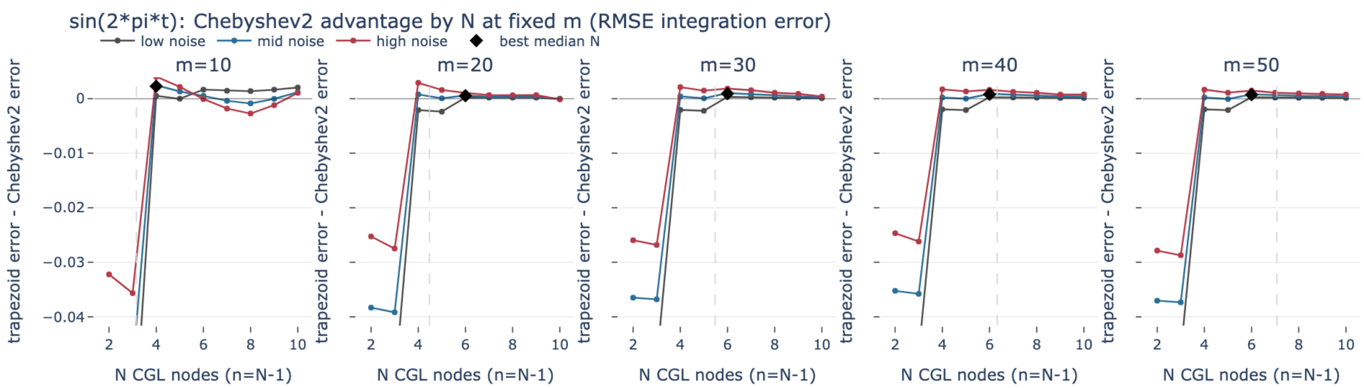

Figure 11: RMSE advantage curves for `sin(2*pi*t)` with Tikhonov (`lambda1 = 0.005`).

sin(2*pi*t): ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	2	4	4	4	75%	100%	100%	0.002278	..@@@@@@@
20	4.47	4	6	6	6	100%	100%	100%	0.0005302	..@@@@@@@
30	5.48	10	6	6	6	75%	100%	100%	0.0009199	..@@@@@@@
40	6.32	2	6	6	6	75%	100%	100%	0.0008021	..@@@@@@@
50	7.07	2	6	6	6	75%	100%	100%	0.0007325	..@@@@@@@

Figure 12: Robust `N` table for `sin(2*pi*t)` with Tikhonov (`lambda1 = 0.005`).

3.2.3 `pi*t^(31/4)`

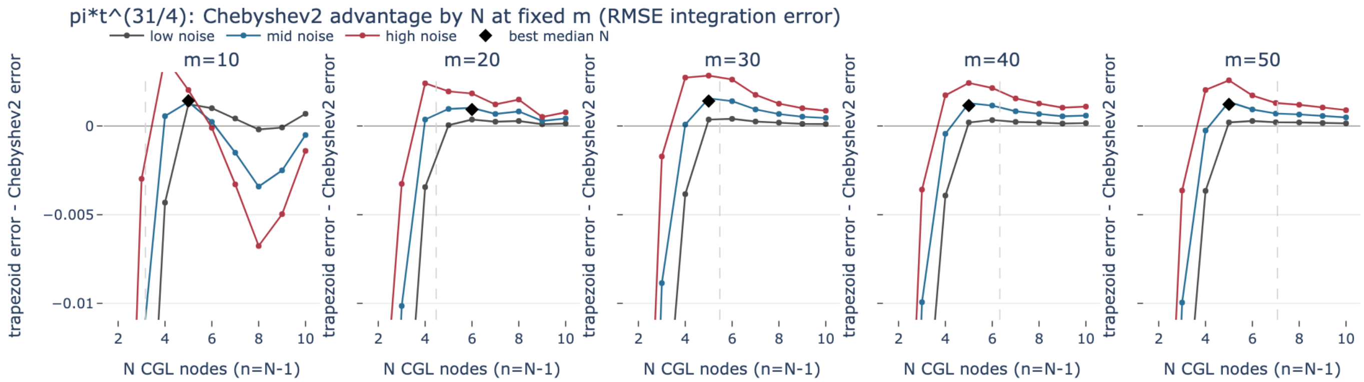

Figure 13: RMSE advantage curves for `pi*t^(31/4)` with Tikhonov (`lambda1 = 0.005`).

pi*t^(31/4): ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	6	5	5	5	100%	100%	100%	0.00142	.#@@@@%@@
20	4.47	3	6	5	6	75%	100%	100%	0.00093	.#@@@@@@@
30	5.48	6	5	5	5	75%	100%	100%	0.001406	.#@@@@@@@
40	6.32	5	5	6	5	50%	100%	100%	0.001152	.#@@@@@@@
50	7.07	10	5	5	5	50%	100%	100%	0.001218	.#@@@@@@@

Figure 14: Robust `N` table for `pi*t^(31/4)` with Tikhonov (`lambda1 = 0.005`).

4 Random Degree-4 Polynomials

The random-polynomial notebook broadens the scalar comparison beyond the three hand-picked analytic functions. Each degree-4 polynomial is defined by fixed random values at five CGL nodes on `[0, 1]`; the same Monte Carlo grid is then used for no Tikhonov and for `lambda1 = 0.005`.

Function	f(0.000)	f(0.146)	f(0.500)	f(0.854)	f(1.000)
random degree-4 polynomial 1	-0.1211	0.3145	0.1258	0.4067	-0.3488
random degree-4 polynomial 2	-0.1641	-0.0227	0.0029	-0.1796	0.199
random degree-4 polynomial 3	-0.4065	0.4601	-0.2177	0.1236	0.423

4.1 No Tikhonov (`lambda1 = 0`)

4.1.1 `random degree-4 polynomial 1`

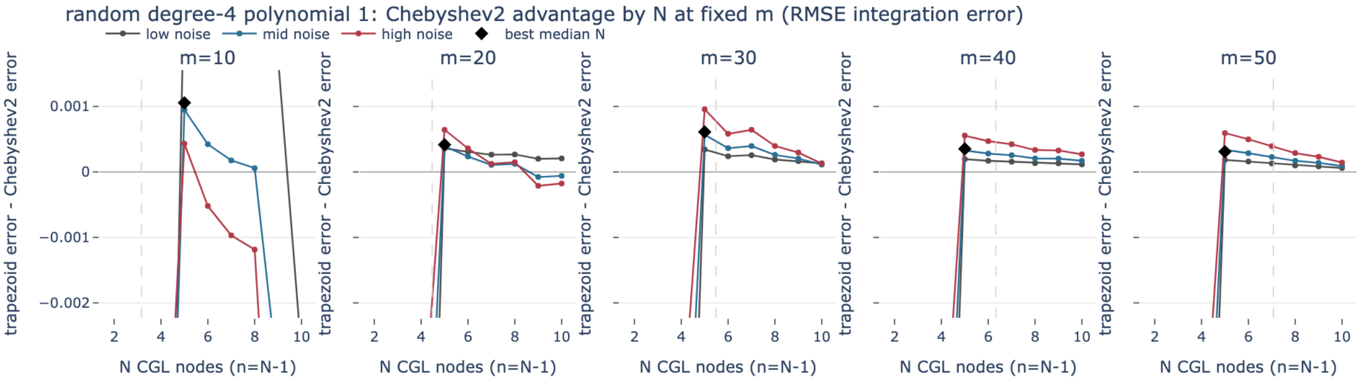

Figure 15: RMSE advantage curves for `random degree-4 polynomial 1` with No Tikhonov (`lambda1 = 0`).

random degree-4 polynomial 1: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	6	5	5	5	100%	100%	100%	0.001055	.+=@@@@#:
20	4.47	6	5	5	5	100%	100%	100%	0.0004163	._:@@@@@@
30	5.48	9	5	5	5	100%	100%	100%	0.0006104	.__@@@@@@
40	6.32	9	5	5	5	100%	100%	100%	0.0003521	.._@@@@@@
50	7.07	8	5	5	5	100%	100%	100%	0.0003103	.._@@@@@@

Figure 16: Robust `N` table for `random degree-4 polynomial 1` with No Tikhonov (`lambda1 = 0`).

4.1.2 `random degree-4 polynomial 2`

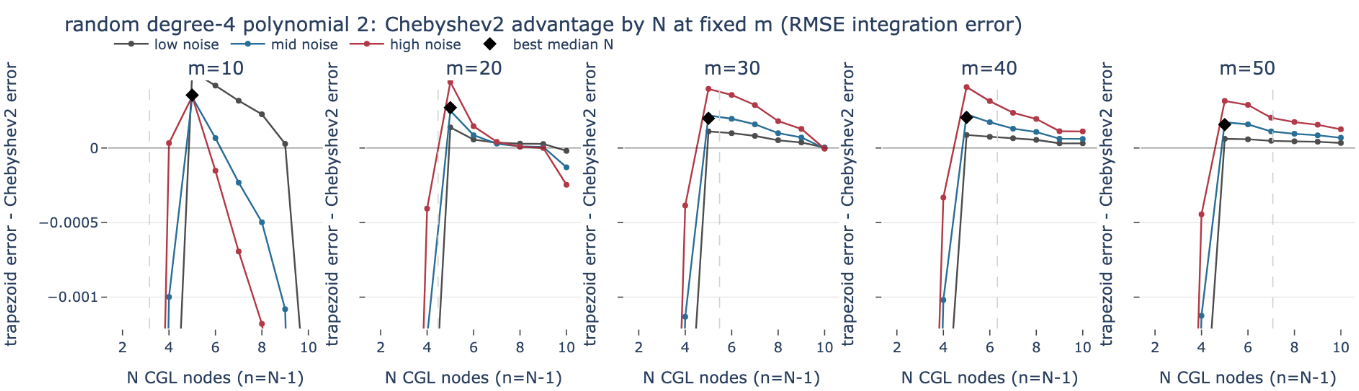

Figure 17: RMSE advantage curves for `random degree-4 polynomial 2` with No Tikhonov (`lambda1 = 0`).

random degree-4 polynomial 2: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	5	5	5	5	100%	100%	100%	0.000355	..%@@@%%=
20	4.47	6	5	5	5	100%	100%	100%	0.0002709	..#@@@@@@
30	5.48	10	5	5	5	100%	100%	100%	0.0001986	..#@@@@@@
40	6.32	6	5	5	5	100%	100%	100%	0.0002049	..%@@@@@@
50	7.07	9	5	5	5	100%	100%	100%	0.0001571	..#@@@@@@

Figure 18: Robust `N` table for `random degree-4 polynomial 2` with No Tikhonov (`lambda1 = 0`).

4.1.3 `random degree-4 polynomial 3`

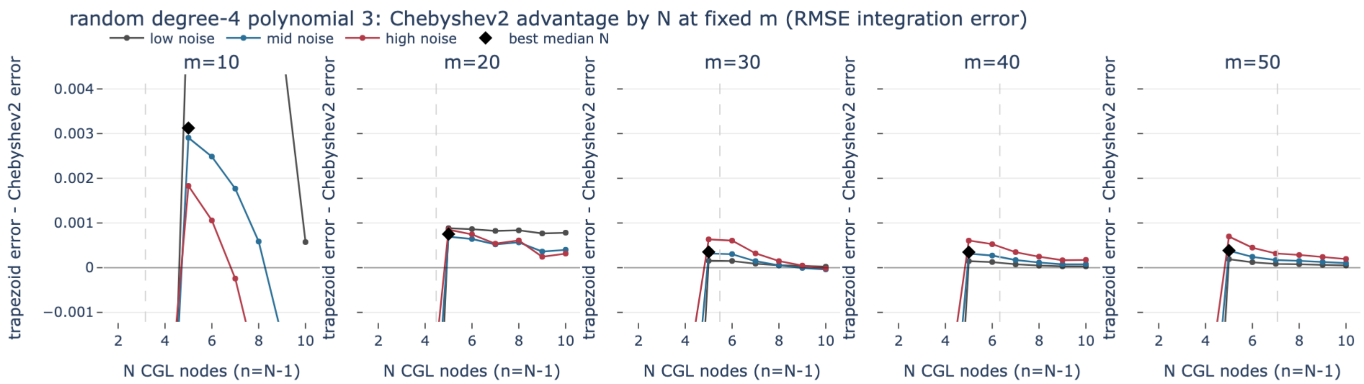

Figure 19: RMSE advantage curves for `random degree-4 polynomial 3` with No Tikhonov (`lambda1 = 0`).

random degree-4 polynomial 3: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	5	5	5	5	100%	100%	100%	0.003125	..#@@@@%+
20	4.47	5	5	5	5	100%	100%	100%	0.0007515	._#@@@@@@
30	5.48	2	5	5	5	100%	100%	100%	0.0003505	.:#@@@@@@
40	6.32	2	5	5	5	100%	100%	100%	0.0003465	.-#@@@@@@
50	7.07	9	5	5	5	100%	100%	100%	0.000384	.-#@@@@@@

Figure 20: Robust `N` table for `random degree-4 polynomial 3` with No Tikhonov (`lambda1 = 0`).

4.2 Tikhonov (`lambda1 = 0.005`)

4.2.1 `random degree-4 polynomial 1`

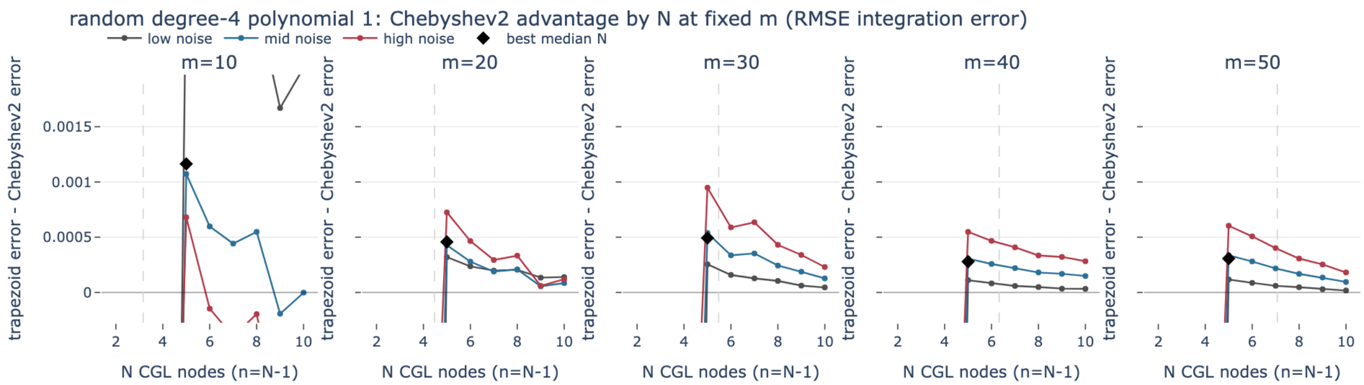

Figure 21: RMSE advantage curves for `random degree-4 polynomial 1` with Tikhonov (`lambda1 = 0.005`).

random degree-4 polynomial 1: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	5	5	5	5	100%	100%	100%	0.001163	.+=@@@@%@
20	4.47	5	5	5	5	100%	100%	100%	0.0004577	._:@@@@@@
30	5.48	7	5	5	7	100%	100%	100%	0.0003219	.__@@@@@@
40	6.32	7	5	5	7	75%	100%	100%	0.0001983	.._@@@@@@
50	7.07	7	5	5	5	75%	100%	100%	0.0003066	.._@@@@@@

Figure 22: Robust `N` table for `random degree-4 polynomial 1` with Tikhonov (`lambda1 = 0.005`).

4.2.2 `random degree-4 polynomial 2`

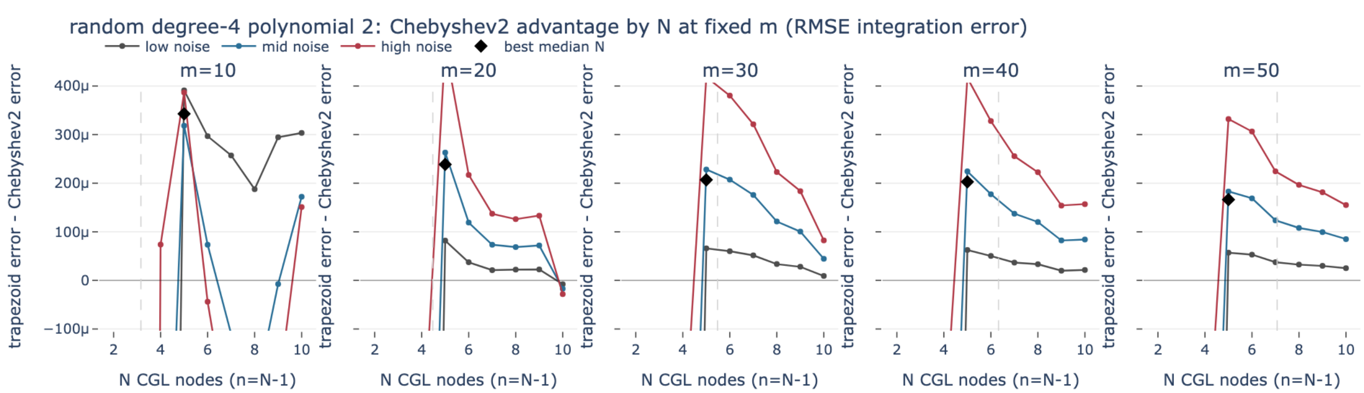

Figure 23: RMSE advantage curves for `random degree-4 polynomial 2` with Tikhonov (`lambda1 = 0.005`).

random degree-4 polynomial 2: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	10	5	5	5	100%	100%	100%	0.0003427	..%@@@@@@
20	4.47	5	5	5	5	75%	100%	100%	0.0002387	..#@@@@@@
30	5.48	7	5	5	5	75%	100%	100%	0.000207	..#@@@@@@
40	6.32	5	5	5	5	75%	100%	100%	0.0002027	..#@@@@@@
50	7.07	9	5	5	5	75%	100%	100%	0.0001663	..#@@@@@@

Figure 24: Robust `N` table for `random degree-4 polynomial 2` with Tikhonov (`lambda1 = 0.005`).

4.2.3 `random degree-4 polynomial 3`

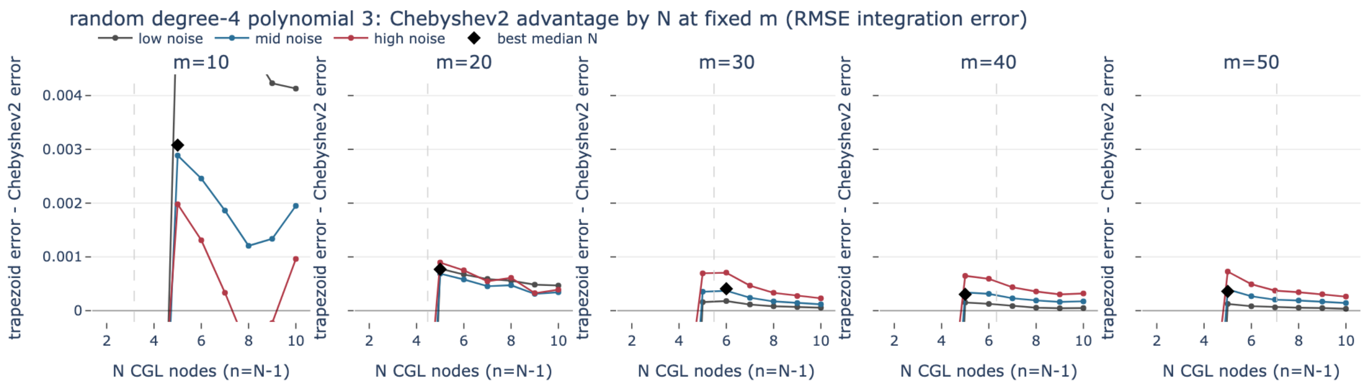

Figure 25: RMSE advantage curves for `random degree-4 polynomial 3` with Tikhonov (`lambda1 = 0.005`).

random degree-4 polynomial 3: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	6	5	5	5	100%	100%	100%	0.003078	..#@@@@@@
20	4.47	5	5	5	5	100%	100%	100%	0.0007689	._#@@@@@@
30	5.48	2	6	5	6	100%	100%	100%	0.0004067	.:#@@@@@@
40	6.32	2	5	5	5	100%	100%	100%	0.0003033	.-#@@@@@@
50	7.07	9	5	5	5	100%	100%	100%	0.000358	.-#@@@@@@

Figure 26: Robust `N` table for `random degree-4 polynomial 3` with Tikhonov (`lambda1 = 0.005`).

5 EuRoC Aggressive 200 ms Snippets

The EuRoC extension changes the question from synthetic scalar functions to IMU-like signals. For each merged EuRoC CSV, the workflow fits one-second `gyro_norm` windows with Chebyshev2 using `N = 50`. It then scores only the center interval `[0.4, 0.6]` with the local aggressive-window diagnostic, so the selected window is aggressive in the 200 ms interval actually used by the quadrature comparison. The resulting CGL node values define the reference scalar function; the Monte Carlo comparison samples only that middle 200 ms interval.

The figure below shows all selected snippets. Points are the raw uniformly sampled IMU values in the selected one-second window, the line is the `N = 50` Chebyshev2 reference fit, and the shaded region is the 200 ms interval used for quadrature.

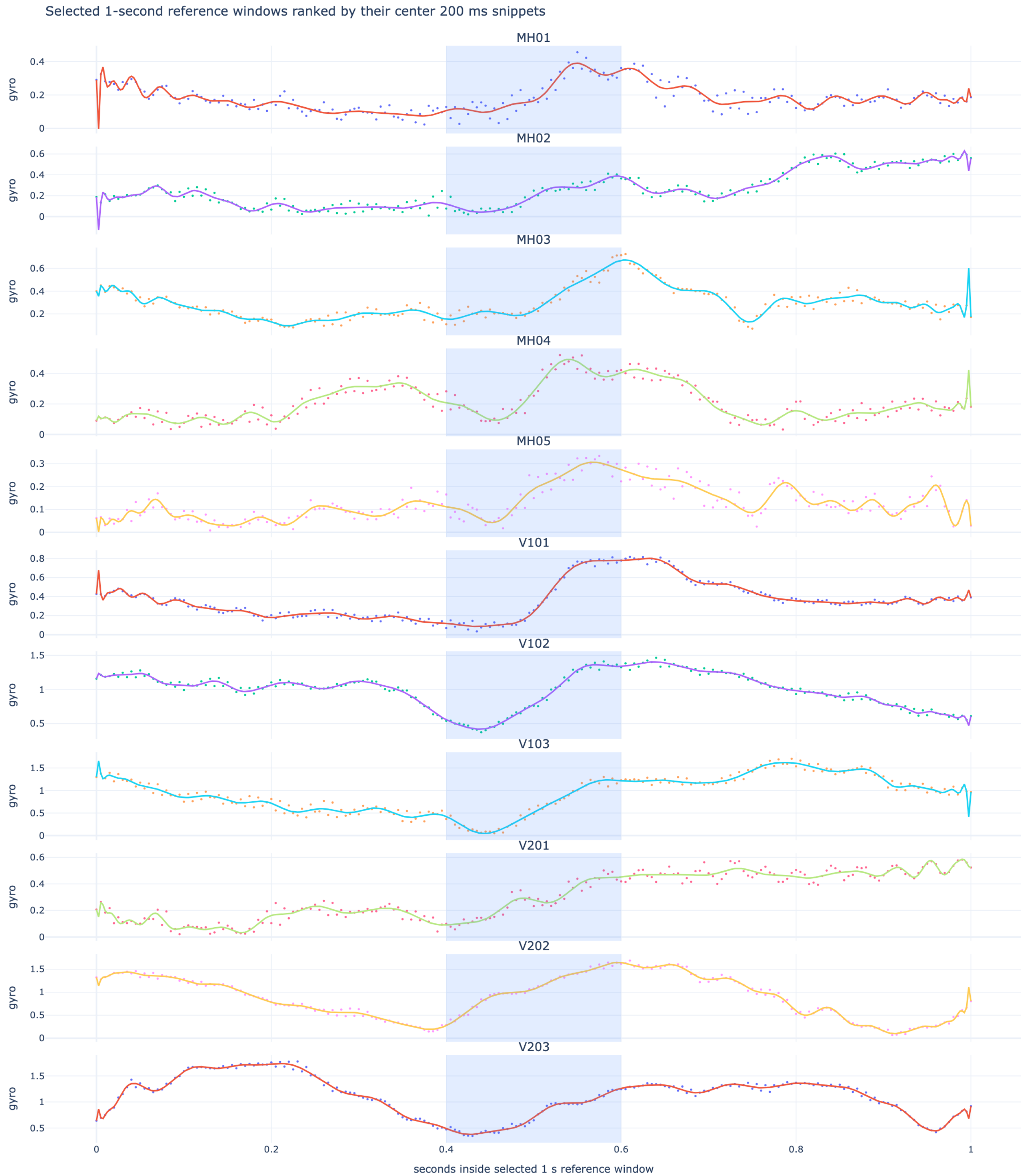

Figure 27: All selected EuRoC aggressive snippets.

Selection metadata:

dataset	window	snippet_s	score	activity	high_ratio	nRMSE
V101	23	23.4-23.6	13.87	1.621	0.8876	0.1391
V103	68	68.4-68.6	9.266	0.8771	0.8588	0.1117

dataset	window	snippet_s	score	activity	high_ratio	nRMSE
V102	69	69.4-69.6	8.143	0.929	0.7507	0.07532
MH02	53	53.4-53.6	7.924	1.2	0.8361	0.2954
MH03	45	45.4-45.6	7.782	0.9836	0.7857	0.1817
MH04	49	49.4-49.6	6.888	1.038	0.7596	0.2433
V202	54	54.4-54.6	6.555	0.9056	0.7245	0.06962
V201	85	85.4-85.6	6.465	0.7833	0.7612	0.1882
MH01	38	38.4-38.6	6.373	1.017	0.7885	0.3344
MH05	16	16.4-16.6	6.302	0.9104	0.7986	0.2625
V203	73	73.4-73.6	5.435	0.6895	0.7481	0.06814

6 EuRoC Decision-oriented Views

The following pages show the same decision views for every selected EuRoC snippet, first without Tikhonov and then with `lambda1 = 0.005`.

6.1 No Tikhonov (`lambda1 = 0`)

6.1.1 `MH01 center-200ms aggressive gyro_norm`

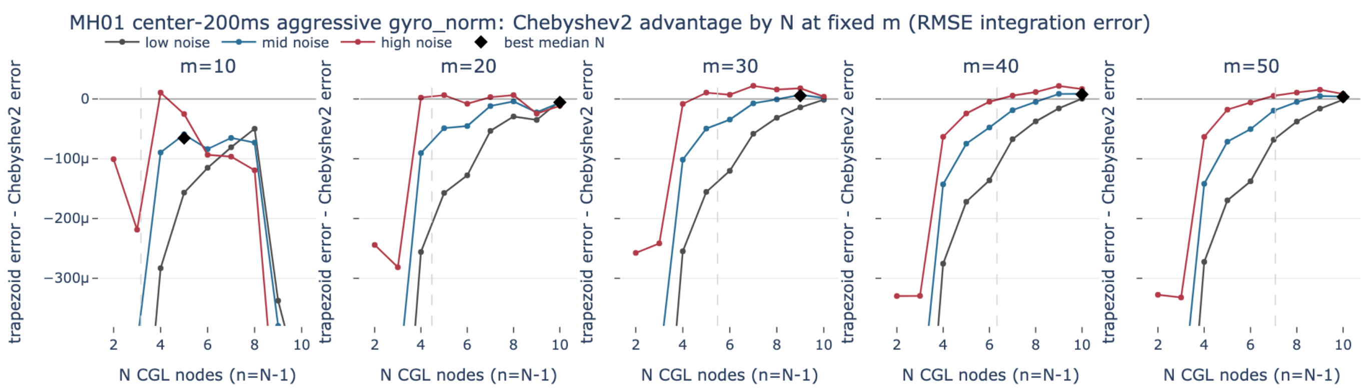

Figure 28: RMSE advantage curves for `MH01 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

MH01 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	2	5	7	4	0%	0%	67%	−0.0001047	**@@@@@*.
20	4.47	2	10	10	8	0%	25%	100%	−5.878e−06	._#%%@@@@
30	5.48	10	9	10	10	25%	100%	100%	1.379e−06	._#%%@@@@
40	6.32	9	10	10	10	50%	100%	100%	7.669e−06	..#%%@@@@
50	7.07	2	10	10	10	50%	100%	100%	3.415e−06	._#%%@@@@

Figure 29: Robust `N` table for `MH01 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

6.1.2 `MH02 center-200ms aggressive gyro_norm`

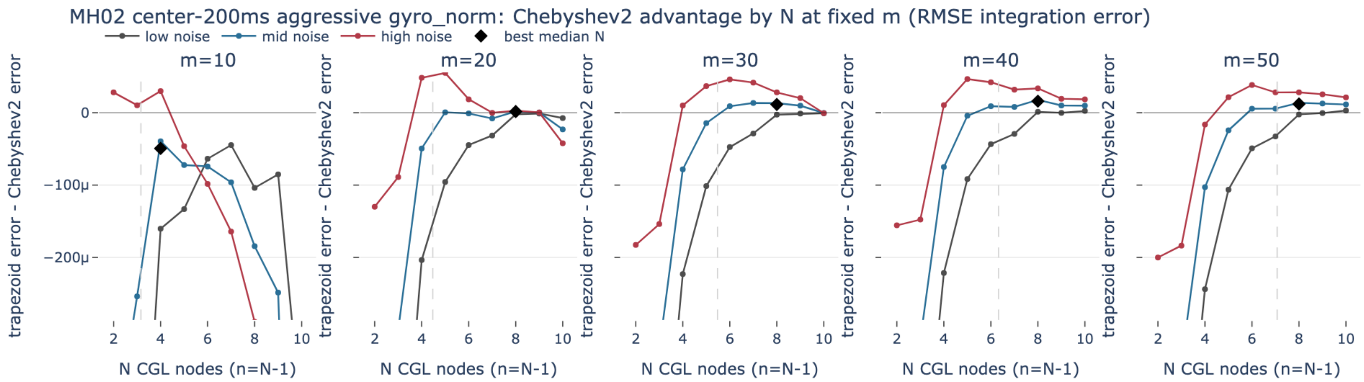

Figure 30: RMSE advantage curves for `MH02 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

MH02 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	4	4	5	4	0%	0%	100%	-4.938e-05	+#@@@@%#.
20	4.47	5	8	8	8	25%	100%	100%	1.559e-06	.-%@@@@@@
30	5.48	10	8	8	8	50%	100%	100%	1.142e-05	.:#%@@@@@
40	6.32	4	8	8	8	50%	100%	100%	1.593e-05	.:#%@@@@@
50	7.07	4	8	10	10	75%	100%	100%	1.026e-05	.:#%@@@@@

Figure 31: Robust `N` table for `MH02 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

6.1.3 `MH03 center-200ms aggressive gyro_norm`

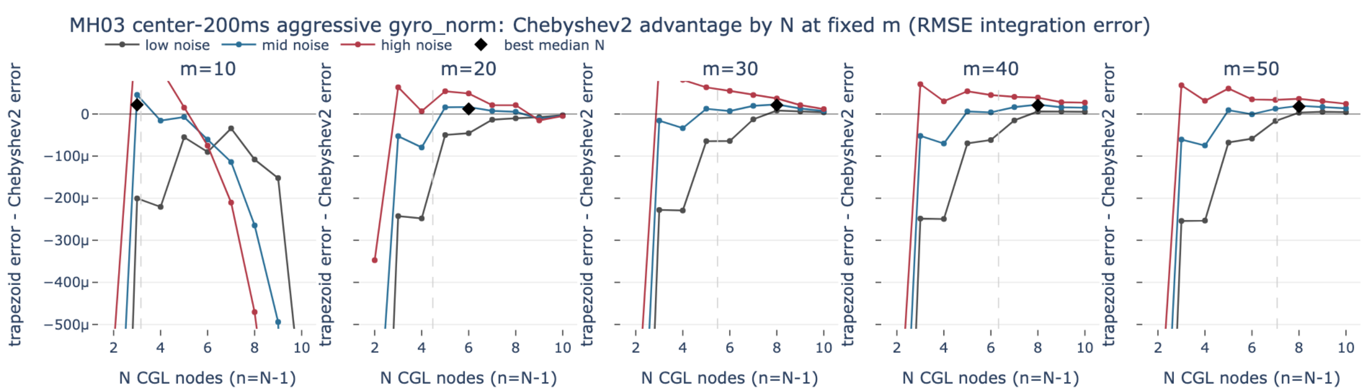

Figure 32: RMSE advantage curves for `MH03 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

MH03 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	4	3	3	3	0%	75%	100%	2.183e-05	−@@@@%%#.
20	4.47	2	6	5	5	0%	75%	100%	1.098e-05	.%%@@@@@@
30	5.48	4	8	8	8	75%	100%	100%	2.11e-05	.@%@@@@@@
40	6.32	4	8	8	8	75%	100%	100%	2.048e-05	.%%@@@@@@
50	7.07	9	8	8	8	50%	100%	100%	1.77e-05	.%%@@@@@@

Figure 33: Robust N table for MH03 center-200ms aggressive gyro_norm with No Tikhonov (lambda1 = 0).

6.1.4 MH04 center-200ms aggressive gyro_norm

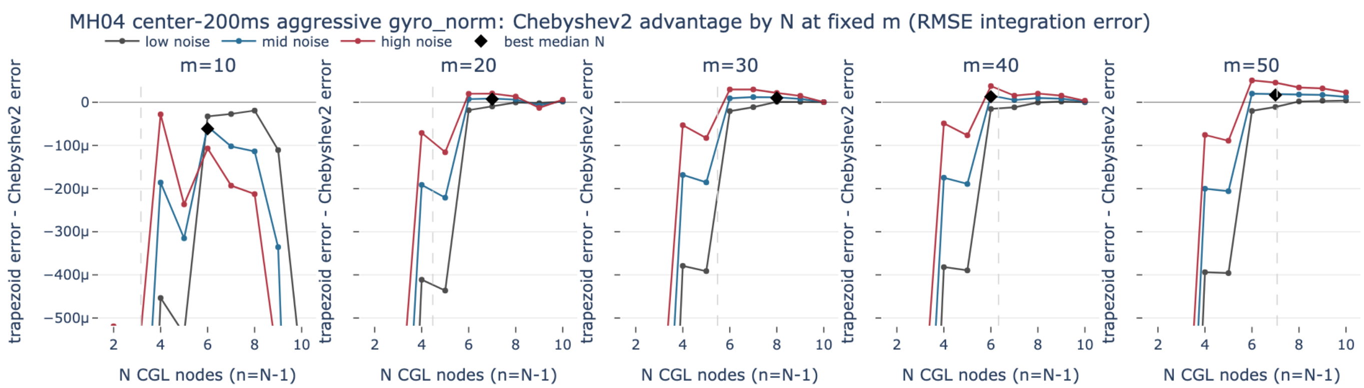

Figure 34: RMSE advantage curves for MH04 center-200ms aggressive gyro_norm with No Tikhonov (lambda1 = 0).

MH04 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	2	6	6	6	0%	0%	0%	−6.136e-05	−−%#@@@%.
20	4.47	2	7	6	7	0%	100%	100%	7.339e-06	..##@@@@@
30	5.48	2	8	8	8	50%	100%	100%	1.027e-05	..%#@@@@@
40	6.32	2	6	6	6	0%	100%	100%	1.295e-05	..%#@@@@@
50	7.07	2	7	6	8	50%	100%	100%	1.614e-05	..##@@@@@

Figure 35: Robust N table for MH04 center-200ms aggressive gyro_norm with No Tikhonov (lambda1 = 0).

6.1.5 `MH05 center-200ms aggressive gyro_norm`

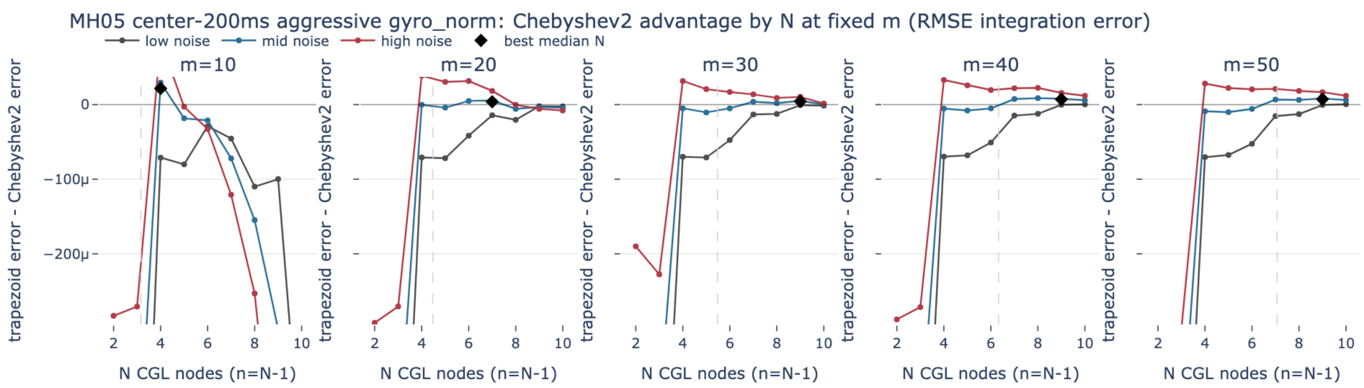

Figure 36: RMSE advantage curves for `MH05 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

MH05 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	4	4	4	4	0%	75%	100%	2.16e-05	=+@@@%%#.
20	4.47	5	7	4	4	0%	50%	100%	-5.794e-06	._@@@@@@@
30	5.48	2	9	9	8	0%	75%	100%	9.685e-07	..@@@@@@@
40	6.32	8	9	9	8	0%	100%	100%	6.811e-06	._@@@@@@@
50	7.07	2	9	9	9	50%	100%	100%	7.325e-06	._@@@@@@@

Figure 37: Robust `N` table for `MH05 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

6.1.6 `V101 center-200ms aggressive gyro_norm`

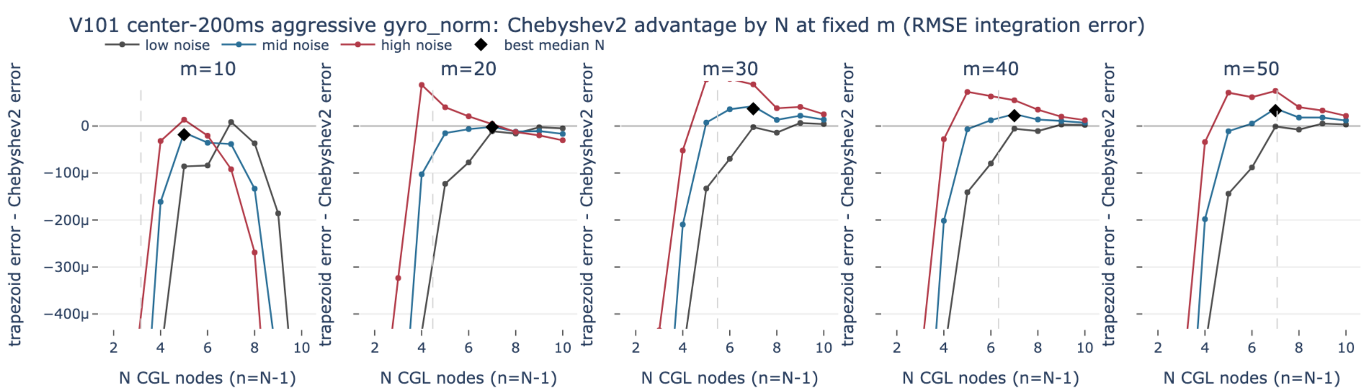

Figure 38: RMSE advantage curves for `V101 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

V101 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	5	5	6	5	0%	25%	100%	−1.834e−05	_+%@@@@#.
20	4.47	2	7	7	7	0%	25%	100%	−2.433e−06	.=%@@@@@@
30	5.48	6	7	7	6	0%	100%	100%	2.71e−05	.=%@@@@@@
40	6.32	2	7	7	7	50%	100%	100%	2.201e−05	.=%@@@@@@
50	7.07	2	7	7	7	50%	100%	100%	3.288e−05	.=%@@@@@@

Figure 39: Robust N table for V101 center-200ms aggressive gyro_norm with No Tikhonov (lambda1 = 0).

6.1.7 V102 center-200ms aggressive gyro_norm

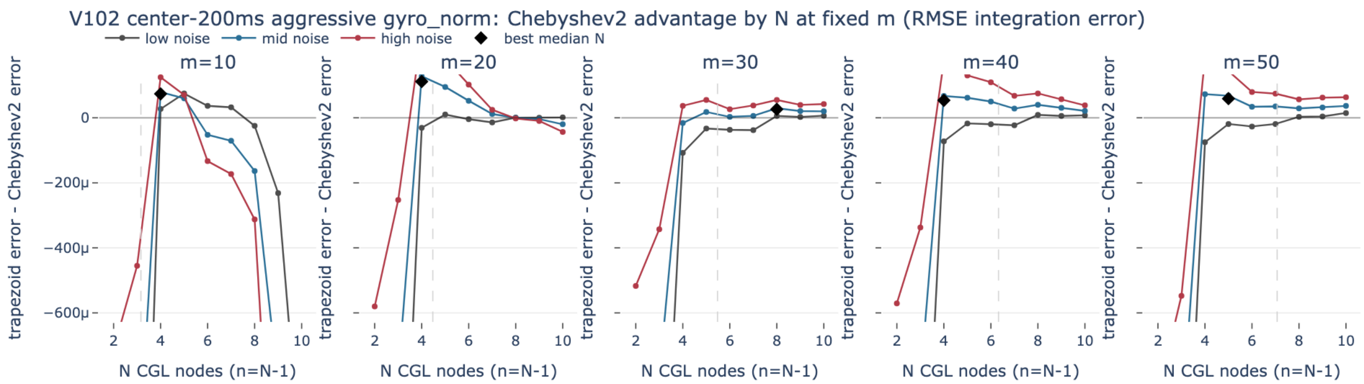

Figure 40: RMSE advantage curves for V102 center-200ms aggressive gyro_norm with No Tikhonov (lambda1 = 0).

V102 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	7	4	5	5	100%	100%	100%	6.363e−05	−+@@@@%#.
20	4.47	4	4	4	4	50%	100%	100%	0.0001115	.−@@@%%%%
30	5.48	7	8	5	8	75%	100%	100%	2.642e−05	.−@@@@@@@
40	6.32	3	4	4	4	0%	100%	100%	5.37e−05	.−@@@@@@@
50	7.07	4	5	5	4	0%	100%	100%	5.839e−05	.−@@@@@@@

Figure 41: Robust N table for V102 center-200ms aggressive gyro_norm with No Tikhonov (lambda1 = 0).

6.1.8 V103 center-200ms aggressive gyro_norm

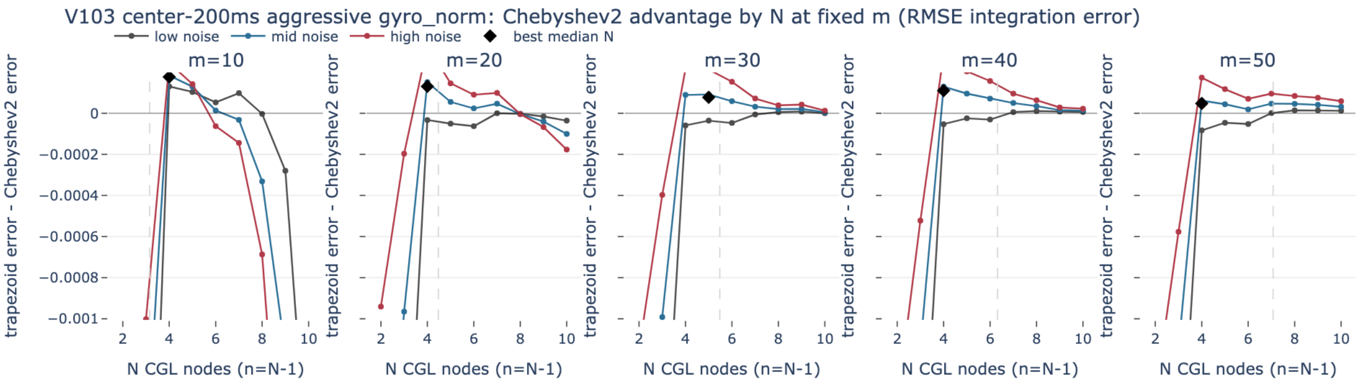

Figure 42: RMSE advantage curves for V103 center-200ms aggressive gyro_norm with No Tikhonov (lambda1 = 0).

V103 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	6	4	4	4	100%	100%	100%	0.0001765	:*@@@@%#.
20	4.47	4	4	4	4	50%	100%	100%	0.0001307	.+@@@@@%%
30	5.48	5	5	4	5	25%	100%	100%	7.73e-05	.+@@@@@@@
40	6.32	3	4	4	4	25%	100%	100%	0.0001101	.=@@@@@@@
50	7.07	7	4	4	7	50%	100%	100%	4.14e-05	.=@@@@@@@

Figure 43: Robust N table for V103 center-200ms aggressive gyro_norm with No Tikhonov (lambda1 = 0).

6.1.9 V201 center-200ms aggressive gyro_norm

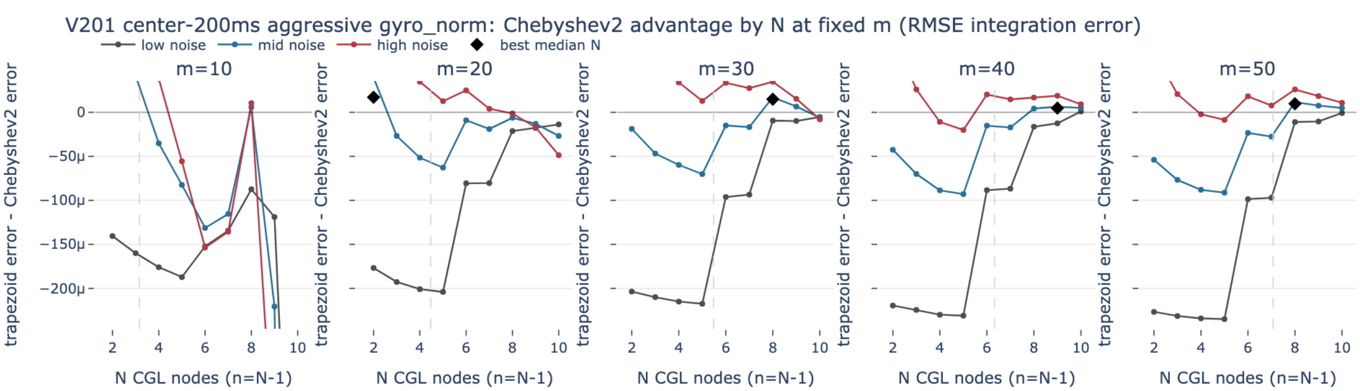

Figure 44: RMSE advantage curves for V201 center-200ms aggressive gyro_norm with No Tikhonov (lambda1 = 0).

V201 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	2	2	2	2	0%	100%	100%	9.226e-05	@@%%%%@#.
20	4.47	2	2	2	2	0%	75%	100%	1.711e-05	@-_.*+#*+
30	5.48	5	8	8	8	25%	100%	100%	1.485e-05	=:_.**@%#
40	6.32	9	9	10	9	0%	100%	100%	4.744e-06	=:..##@@@
50	7.07	6	8	8	8	25%	100%	100%	9.781e-06	-_..**@@@

Figure 45: Robust `N` table for `V201 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

6.1.10 `V202 center-200ms aggressive gyro_norm`

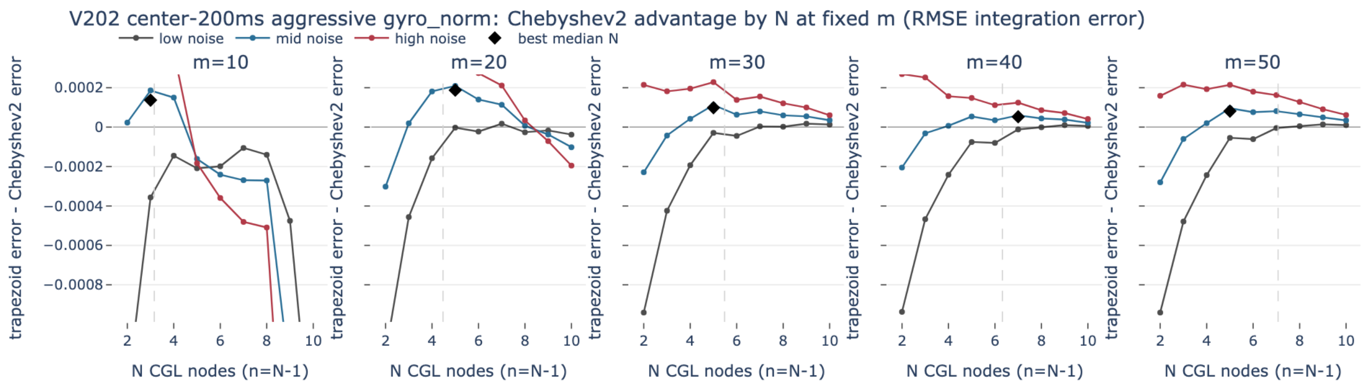

Figure 46: RMSE advantage curves for `V202 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

V202 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	4	3	3	3	0%	75%	100%	0.0001369	@@@%%%%#.
20	4.47	4	5	5	4	0%	100%	100%	0.0001468	.*%@%%**+
30	5.48	5	5	5	5	50%	100%	100%	9.855e-05	.+#@%%%%#
40	6.32	7	7	5	7	50%	100%	100%	5.253e-05	.*#@%@@@%
50	7.07	7	5	5	5	25%	100%	100%	8.105e-05	.+#@@@@%%

Figure 47: Robust `N` table for `V202 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

6.1.11 `V203 center-200ms aggressive gyro_norm`

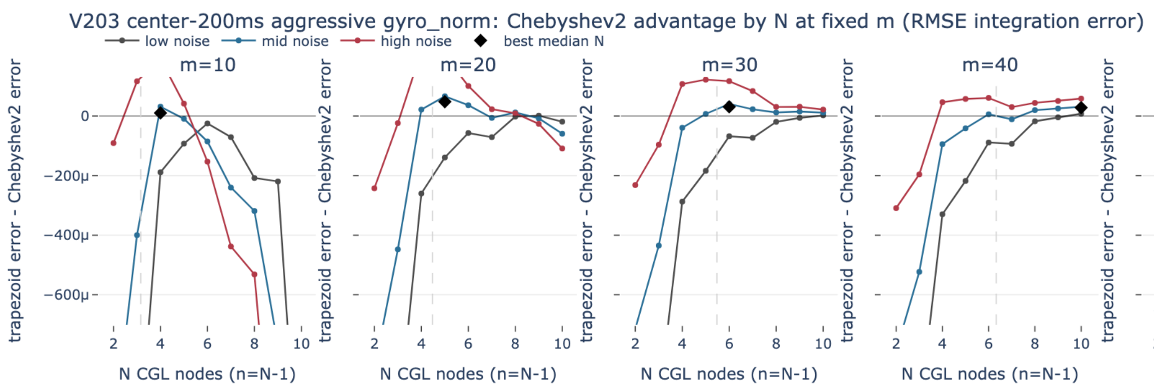

Figure 48: RMSE advantage curves for `V203 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

V203 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	4	4	5	4	0%	75%	100%	1.02e-05	*%@@@%%#.
20	4.47	6	5	5	5	0%	75%	100%	4.799e-05	.=@@@%@%%
30	5.48	6	6	6	6	0%	100%	100%	3.06e-05	.-%@@@@@@
40	6.32	6	10	8	10	75%	100%	100%	2.798e-05	.-%%@@@@@
50	7.07	7	9	8	8	25%	100%	100%	2.922e-05	.-%%@@@@@

Figure 49: Robust `N` table for `V203 center-200ms aggressive gyro_norm` with No Tikhonov (`lambda1 = 0`).

6.2 Tikhonov (`lambda1 = 0.005`)

6.2.1 `MH01 center-200ms aggressive gyro_norm`

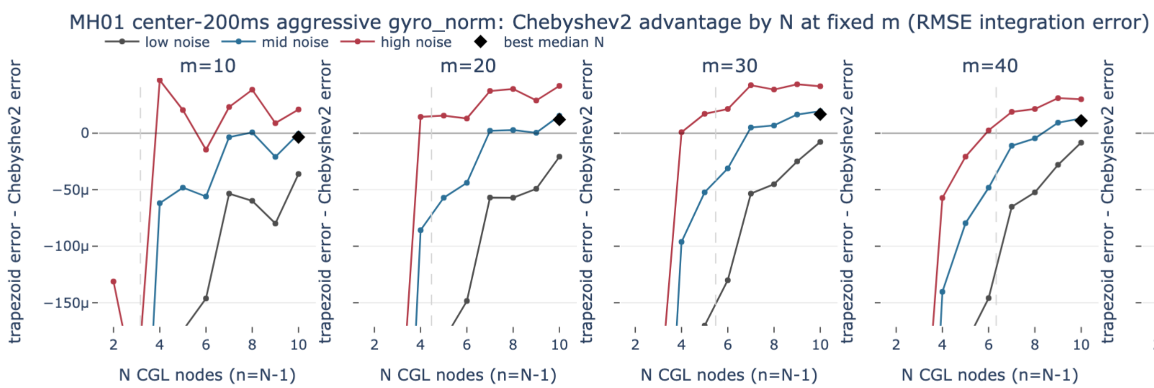

Figure 50: RMSE advantage curves for `MH01 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

MH01 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	2	10	10	8	0%	50%	100%	−4.441e−06	..%%%@@@@
20	4.47	2	10	10	10	0%	100%	100%	1.205e−05	._#%%@@@@
30	5.48	9	10	10	10	50%	100%	100%	1.678e−05	._#%%@@@@
40	6.32	9	10	10	9	0%	75%	100%	6.305e−06	..#%%@@@@
50	7.07	2	10	10	10	50%	100%	100%	1.117e−05	._#%%@@@@

Figure 51: Robust `N` table for `MH01 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

6.2.2 `MH02 center-200ms aggressive gyro_norm`

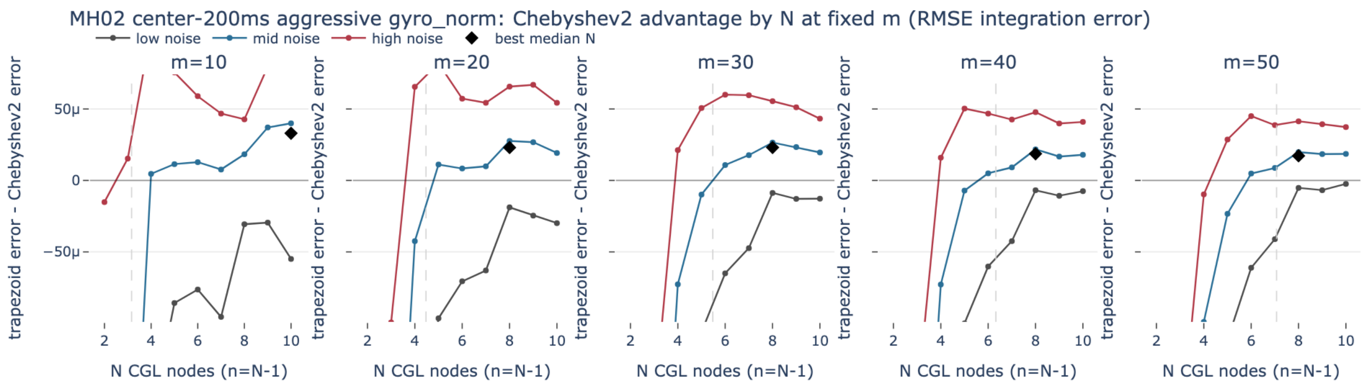

Figure 52: RMSE advantage curves for `MH02 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

MH02 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	9	10	10	10	0%	100%	100%	3.302e−05	.=%@@@@@@
20	4.47	6	8	8	8	25%	100%	100%	2.3e−05	.−%@@@@@@
30	5.48	7	8	8	8	50%	100%	100%	2.306e−05	.:#%@@@@@
40	6.32	6	8	8	8	50%	100%	100%	1.869e−05	.:#%@@@@@
50	7.07	4	8	10	10	50%	100%	100%	1.637e−05	.:#%@@@@@

Figure 53: Robust `N` table for `MH02 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

6.2.3 MH03 center-200ms aggressive gyro_norm

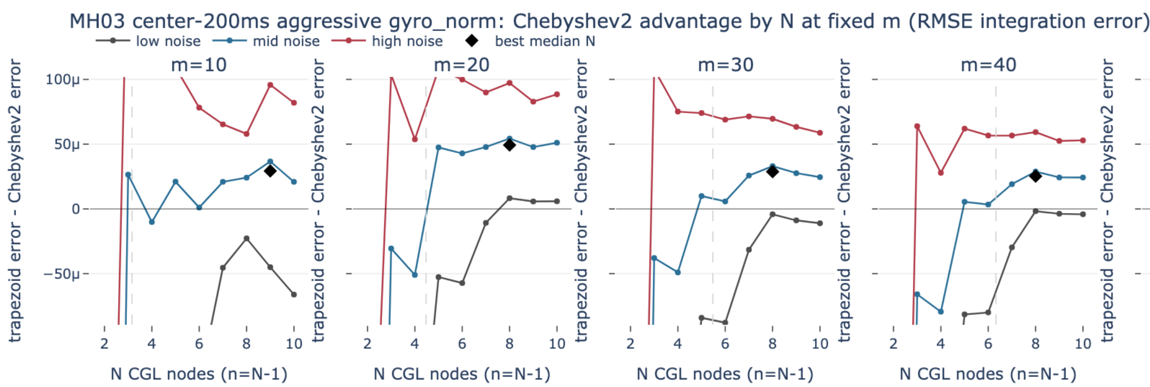

Figure 54: RMSE advantage curves for `MH03 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

MH03 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	4	9	5	9	0%	100%	100%	2.933e-05	.@@@@@@@@
20	4.47	2	8	5	5	0%	100%	100%	3.95e-05	.%%@@@@@@
30	5.48	4	8	8	8	50%	100%	100%	2.878e-05	.%%@@@@@@
40	6.32	3	8	8	8	50%	100%	100%	2.53e-05	.%%@@@@@@
50	7.07	9	8	8	9	50%	100%	100%	2.072e-05	.%%@@@@@@

Figure 55: Robust `N` table for `MH03 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

6.2.4 MH04 center-200ms aggressive gyro_norm

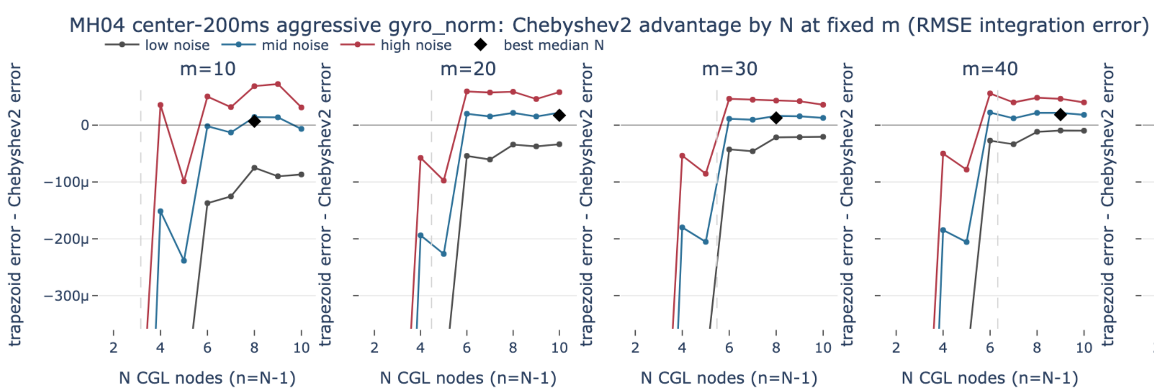

Figure 56: RMSE advantage curves for `MH04 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

MH04 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	2	8	6	6	0%	50%	100%	-1.005e-05	..%#@@@@@
20	4.47	2	10	6	8	0%	100%	100%	1.689e-05	..##@@@@@
30	5.48	2	8	8	8	0%	100%	100%	1.276e-05	..##@@@@@
40	6.32	2	9	8	8	25%	100%	100%	1.842e-05	..##@@@@@
50	7.07	10	9	9	9	25%	100%	100%	1.584e-05	..##@@@@@

Figure 57: Robust `N` table for `MH04 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

6.2.5 `MH05 center-200ms aggressive gyro_norm`

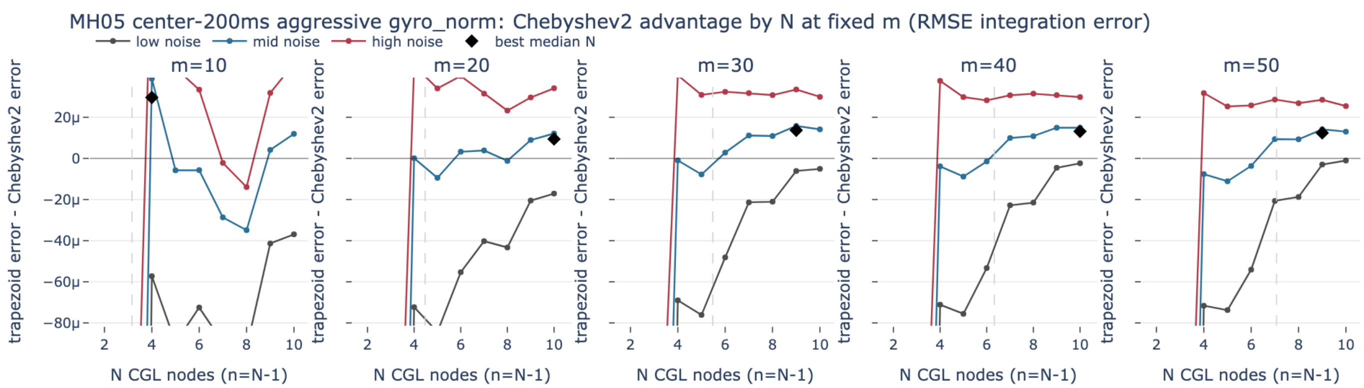

Figure 58: RMSE advantage curves for `MH05 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

MH05 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	3	4	4	4	0%	100%	100%	2.966e-05	._@%@%%@@
20	4.47	5	10	10	10	0%	100%	100%	9.424e-06	._@@@@@@@
30	5.48	2	9	10	10	50%	100%	100%	1.224e-05	..@@@@@@@
40	6.32	8	10	10	8	0%	100%	100%	8.168e-06	._@@@@@@@
50	7.07	2	9	9	9	50%	100%	100%	1.241e-05	._@@@@@@@

Figure 59: Robust `N` table for `MH05 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

6.2.6 V101 center-200ms aggressive gyro_norm

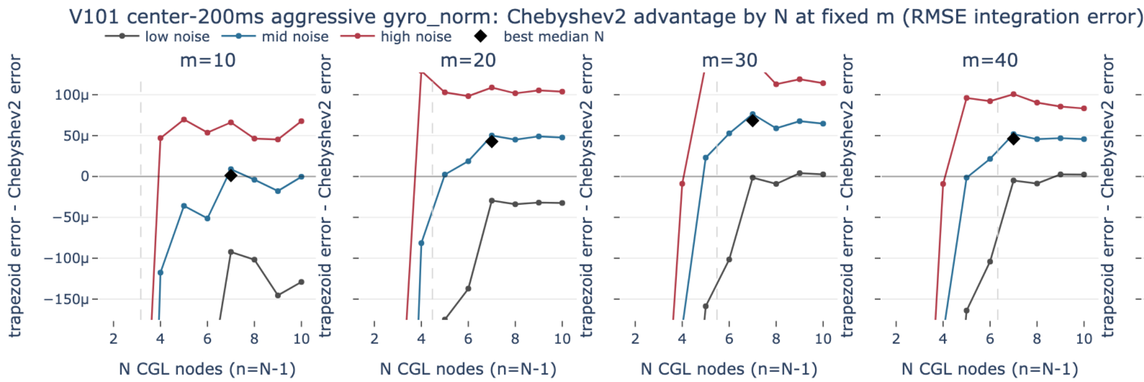

Figure 60: RMSE advantage curves for `V101 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

V101 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	8	7	10	10	0%	50%	100%	-9.77e-06	.=%@@@@@@
20	4.47	2	7	9	9	25%	100%	100%	4.2e-05	.=%@@@@@@
30	5.48	5	7	7	7	50%	100%	100%	6.829e-05	.=%@@@@@@
40	6.32	2	7	7	7	50%	100%	100%	4.602e-05	.=%@@@@@@
50	7.07	2	7	7	9	50%	100%	100%	3.457e-05	.=%@@@@@@

Figure 61: Robust `N` table for `V101 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

6.2.7 V102 center-200ms aggressive gyro_norm

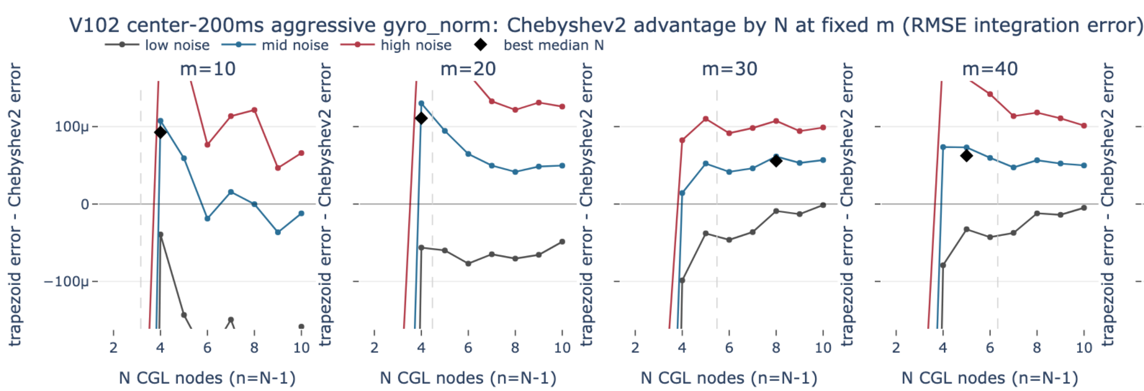

Figure 62: RMSE advantage curves for `V102 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

V102 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	7	4	4	5	0%	75%	100%	4.195e-05	.-@@@@@%@
20	4.47	3	4	4	4	25%	100%	100%	0.0001109	.-@@@@@@@
30	5.48	7	8	8	8	50%	100%	100%	5.564e-05	.-@@@@@@@
40	6.32	4	5	4	4	0%	100%	100%	5.873e-05	.-@@@@@@@
50	7.07	3	5	5	9	0%	100%	100%	2.073e-05	.-@@@@@@@

Figure 63: Robust `N` table for `V102 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

6.2.8 `V103 center-200ms aggressive gyro_norm`

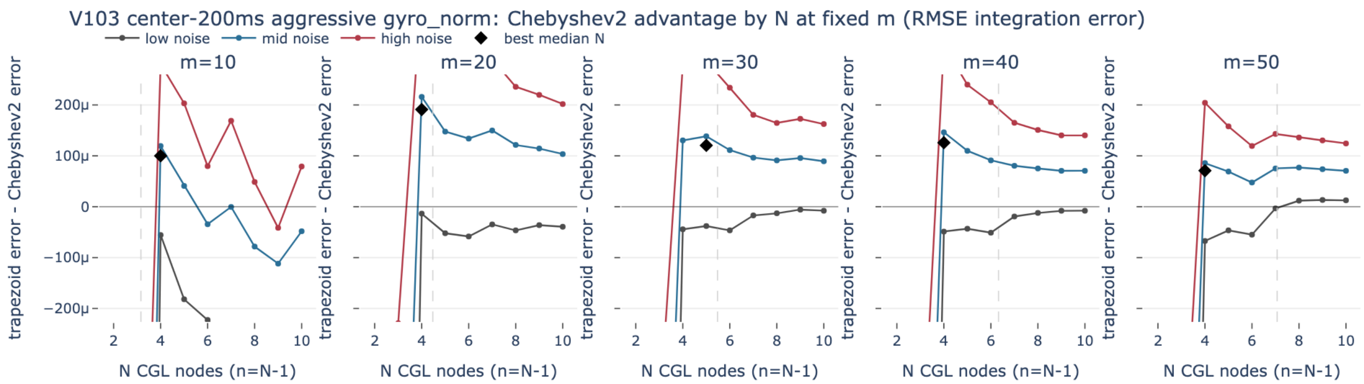

Figure 64: RMSE advantage curves for `V103 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

V103 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	5	4	4	5	0%	75%	100%	2.01e-05	.=@@@@@@@
20	4.47	4	4	4	4	50%	100%	100%	0.0001911	.+@@@@@@@
30	5.48	5	5	4	5	50%	100%	100%	0.0001206	.+@@@@@@@
40	6.32	4	4	4	4	50%	100%	100%	0.0001259	.=@@@@@@@
50	7.07	10	4	4	4	0%	100%	100%	7.089e-05	.=@@@@@@@

Figure 65: Robust `N` table for `V103 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

6.2.9 `V201 center-200ms aggressive gyro_norm`

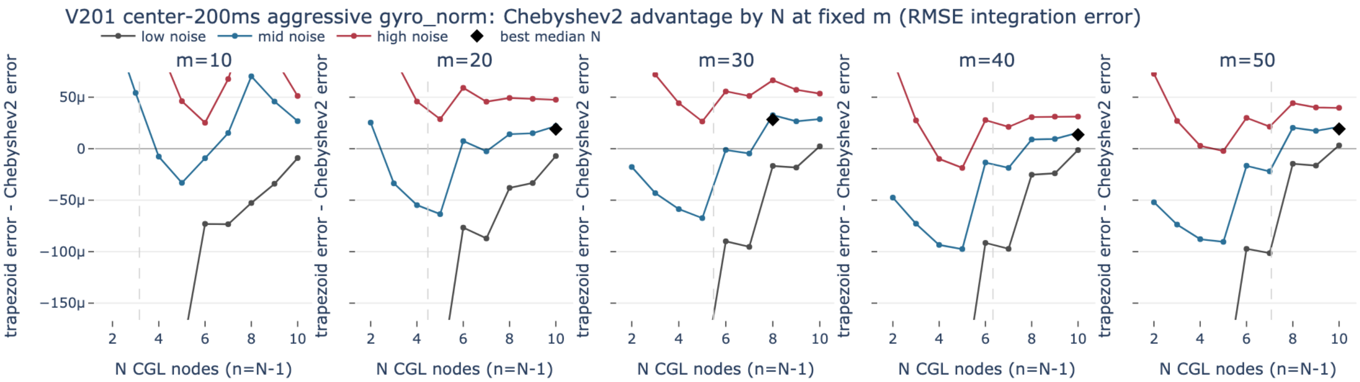

Figure 66: RMSE advantage curves for `V201 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

V201 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	7	2	8	2	0%	75%	100%	8.906e-05	@+_.:=#*+
20	4.47	2	10	10	10	50%	100%	100%	1.899e-05	#:_.#*%%@
30	5.48	5	8	10	8	25%	100%	100%	2.827e-05	=:_.**@@@
40	6.32	10	10	10	10	50%	100%	100%	1.36e-05	-:..#*%%@
50	7.07	5	10	10	8	25%	100%	100%	1.733e-05	-_..**@@@

Figure 67: Robust `N` table for `V201 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

6.2.10 `V202 center-200ms aggressive gyro_norm`

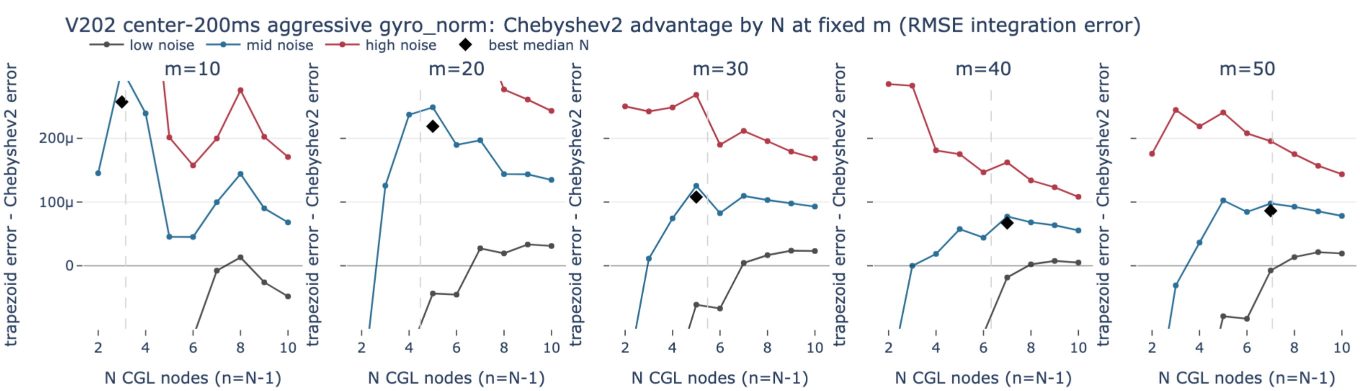

Figure 68: RMSE advantage curves for `V202 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

V202 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	4	3	3	3	25%	100%	100%	0.0002572	.@#..:=:_
20	4.47	3	5	5	4	25%	100%	100%	0.0001981	.#@@%%###
30	5.48	5	5	5	5	25%	100%	100%	0.0001079	.*%@%@@@%
40	6.32	7	7	8	8	50%	100%	100%	6.061e-05	.*#%%@@@%
50	7.07	7	7	9	7	50%	100%	100%	8.626e-05	.*#@@@@@@

Figure 69: Robust `N` table for `V202 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

6.2.11 `V203 center-200ms aggressive gyro_norm`

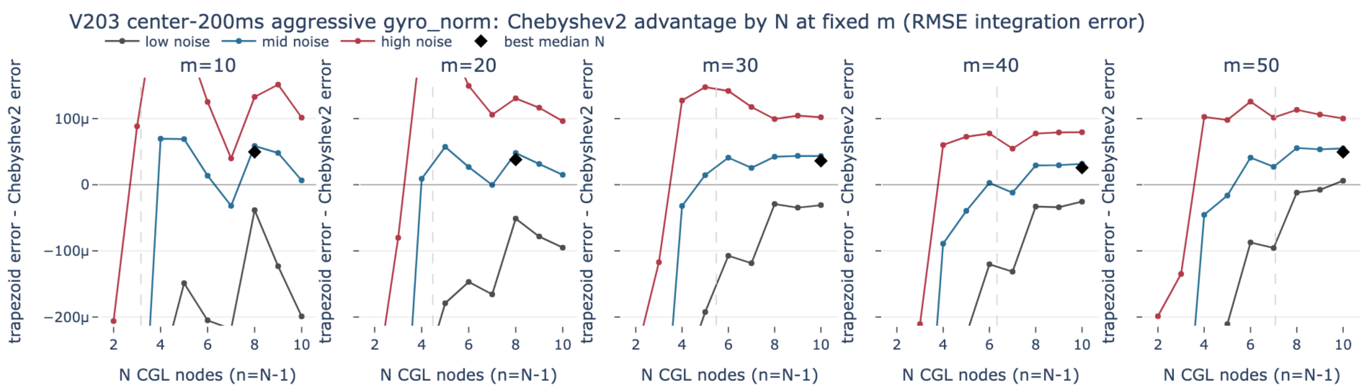

Figure 70: RMSE advantage curves for `V203 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).

V203 center-200ms aggressive gyro_norm: ideal N by metric and robust N (RMSE sparkline is N ascending)

m	sqrt(m)	best final N	best RMSE N	best max N	robust N	low win	med win	high win	RMSE advantage	RMSE advantage over N
10	3.16	5	8	8	5	0%	75%	100%	4.856e-05	.+@@@%@@@
20	4.47	5	8	8	8	0%	100%	100%	3.796e-05	.=@@@@@@@
30	5.48	5	10	8	8	25%	100%	100%	3.54e-05	.-%@@@@@@
40	6.32	5	10	10	10	0%	100%	100%	2.579e-05	.-%%@@@@@
50	7.07	7	10	10	10	50%	100%	100%	4.943e-05	.-%%@@@@@

Figure 71: Robust `N` table for `V203 center-200ms aggressive gyro_norm` with Tikhonov (`lambda1 = 0.005`).